

**2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples**

### **Aim**

To implement the **Candidate-Elimination algorithm** using training examples stored in a CSV file and obtain the set of hypotheses consistent with the training data.

### **Software Required**

- Python 3.x
- Pandas
- Jupyter Notebook / Google Colab

### **Procedure**

1. Store the training examples in a CSV file.
2. Read the CSV file using Pandas.
3. Initialize the Specific boundary with the first positive example.
4. Initialize the General boundary with maximally general hypotheses.
5. Process positive and negative examples.
6. Update the Specific and General boundaries.
7. Display the final version space boundaries.

### **Code**

```
import pandas as pd

data = pd.DataFrame({
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'AirTemp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Water': ['Warm', 'Warm', 'Warm', 'Cool'],
    'Forecast': ['Same', 'Same', 'Change', 'Change'],
    'EnjoySport': ['Yes', 'Yes', 'No', 'Yes']
})

attributes = list(data.columns[:-1])
```

```

S = ['0'] * len(attributes)
G = [['?'] * len(attributes)]
for _, row in data.iterrows():
    example = list(row[:-1])
    target = row.iloc[-1]
    if target == 'Yes':
        for i in range(len(attributes)):
            if S[i] == '0':
                S[i] = example[i]
            elif S[i] != example[i]:
                S[i] = '?'
        G = [g for g in G if all(
            g[i] == '?' or g[i] == example[i]
            for i in range(len(attributes))
        )]
    else:
        new_G = []
        for g in G:
            for i in range(len(attributes)):
                if g[i] == '?' and S[i] != example[i]:
                    new_h = g.copy()
                    new_h[i] = S[i]
                    new_G.append(new_h)
        G = new_G
print("Specific Boundary:")
print(S)
print("\nGeneral Boundary:")
for g in G:
    print(g)

```

### Output:

```
print(g)
```

```
... Specific Boundary:  
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

```
General Boundary:  
['Sunny', '?', '?', '?', '?', '?']  
['?', 'Warm', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']
```

### Result

Thus, the **Candidate-Elimination algorithm** was successfully implemented and the consistent hypothesis boundaries were obtained.